MIL-W-80214B
22 March 1985
SUPERSEDING
MIL-W-80214A
17 November 1976

MILITARY SPECIFICATION

WELDING MACHINES, ARC, DC, TRANSFORMER-RECTIFIER, PULSATING CURRENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE .

1.1 Scope. This specification covers pulsating current, constant potential transformer-rectifier! direct current arc welding machinea with equipment and accessories specified, herein.

1.2 Classification. This specification covers welding machines of the following types and sizes, as specified (see 6.2.1).

Type I - Machines with 1 to not less than 30 pulses per second.

Size 200 - 200 amperage
Size 300 - 300 amperage
Size 450 - 450 amperage
Size 600 - 600 amperage

Type 11 - Machines with steady state, (no pulsing) 60 pulses and not less than 120 pulses per second.

Size 300 - 300 amperage
Size 450 - 450 amperage

Type III - Machines with variable pulses from 50 or less to not less than 250 pulses per second

Size 250 - 250 amperage
Size 350 - 350 amperage
Size 450 - 450 amperage

Beneficial comments (recommendations, additions, deletion8) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3431

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Type IV - Machines with steady stated and with variable pulses from 45 or less to not lees than 180.

Size 300 - 300 amperages
Size 350 - 350 amperages
Size 400 - 400 amperages
Size 450 - 450 amperages
Size 500 - 500 amperages

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-w-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition

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functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/NEMA EW 1 - Electric Arc-Welding Apparatus.

ANSI/NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Design. The welding machine shall be new and one of the manufacturer's current models. The machine shall be of the transformer-rectifier type, solid state design for delivering constant voltage direct current, steady state and pulsed current output in accordance with the requirements specified herein. The machine shall be capable of operating at 100 percent duty cycle and conform to NEMA Standard EW-1. The machine shall include all components, parts, and features necessary to ❑eet the performance require-❑ents specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

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3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (S1) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to S1 units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (S1) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the S1 system of measurements.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The welding machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for a high degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.5 Temperature rise. The maximum allowable temperature rise of various parts of the machines for the given class of insulation system used shall be in accordance with NEMA EW-I. The operating temperature of the rectifier under maximum rated load conditions, shall not exceed a value which would result in injury to the rectifier or adversely affect the performance of the welding machines.

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3.2.6 Portability. The welding machines shall be portable, and shall be base mounted. ” Running gear, if required (see 6.2.1), shall be as specified by the procuring act<vity. The welding machines shall be provided with a lifting device conveniently located for handling by crane or hoist, and shall be capable of supporting 2-1/2 times the weight of the machine. The center of gravity of the entire unit shall be as low as is consistent with economical design. The welding machine shall also be designed to be lifted by forklift truck.

3.2.7 Environmental protection. The machine shall be designed and constructed that, under the operating, service, transportation and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state or local statutes in effect at point of the installation.

3.2.8 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts.

3.2.9 Interchangeability. ‘To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification to the part or the machine.

3.3 Construction. The welding machine shall “be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins , sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the

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condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting,. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color, or one coat of powered epoxy finish color which does not require primer.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Accessibility. The welding machines shall be provided with removable panels or doors for access to terminal boards and other parts which may require replacement or adjustment. Components within the cabinet shall be arranged so that they may be removed or replaced without undue interference from, or removal of, or injury to, other components and wirings.

3.4 Components. The welding machines shall consist of, but not be limited to, enclosure, transformer, rectifier, controls, receptacles and instruments.

3.4.1 Enclosure. The sheet metal enclosure shall be dripproof, so designed and constructed that liquid or solid particles falling on the enclosure from above, at angles up to 15 degrees from the vertical, shall not interfere with the satisfactory operation of the equipment. The welding power terminals shall be recessed or otherwise protected against accidental contact by personnel. All intake and exhaust openings shall be screened or shall be small enough to prevent entrance of trash or rodents.

3.4.2 Transformer. The welding transformer(s) shall be designed to operate from a single or three phase primary power input line. The primary and secondary coils shall be insulated from each other. Insulation of the transformer coils shall be class B, F, or H material in accordance with requirements of NEMA EW-1. Temperature rise of the windings shall not exceed the limits of the applicable class of insulation used as specified in NEMA EW-1. Insulation distances and clearances shall be as liberal as is economically practicable in order to minimize the need for frequent cleaning. The transformer(s) shall be rigidly mounted to the frame of the enclosure.

3.4.2.1 Overtemperature protection. Transformer coil windings shall be protected by a suitable thermal device or devices. The device shall operate to prevent the transformer from reaching injurious temperature by de-energizing or reducing the output of the machine. A combination overtemperature and overload device may be used in lieu of separate controls for overtemperature and overload.

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3.4.3 Rectifier. "he rectifiers shall be either silicon diodes, or other solid state devices such as SCR'S, of suitable size and design for the intended service. Temperature rise of silicon diodes shall not exceed 110 degrees C when operated at rated output and duty cycle.

3.4.4 Controls. The machine shall have controls for all functions, including peak current, peak current time, background voltage, pulse frequency, switch for selecting each mode of operation for the respective type machine, switches for power on-off, remote contactor, control or provisions for remote contactor control, remote output control, power on-off lights. Machines with variable pulse output shall provide for selecting any desired output within the pulse range. Combination control may be provided for multiple functions, if this method is standard with the manufacturer. Al 1 controls, except the wire feed controls or remote controls shall be mounted on the front panel and designed for easy operation by a gloved hand. Al 1 controls shall be permanently and legibly marked. A remote control station to change the welding voltage shall be furnished with a 7.6 meter (25 feet) long connecting cable.

3.4.5 Receptacles. The machine shall have receptacles mounted on the instrument control panel from remote output control, remote contactor control and a 115-volt AC wire feeder power supply. A duplex receptacle shall be provided to supply a minimum of 15 amperes at 115-volts AC for operation of accessory equipment. The remote control receptacles shall provide for twist lock plug connections.

3.4.6 Instruments . The welding machine shall have voltmeters and ammeters or a voltmeter, if digital, mounted on the front panel. The voltmeters shall indicate open circuit voltage and welding voltage. When one voltmeter is used, voltage indications shall be accomplished by use of a ' multiposition switch properly marked. The welding machines ❑ay have separate voltmeters to furnish the above indications. The meters shall have an accuracy of +2 percent on a full scale reading and when in the average position the met~rs shall, read the average value.

3.4.7 Power supply. Unless otherwise specified (see 6.2.1), welding machines shall be designed to operate from a three phase, 60 Hz, 230/460 volt primary power source and shall be connected for use on 3 phase, 460 volts.

3.4.7.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings and shall conform to NEMA-MG 1.

3.4.7.2 Solid state components. Solid state design shall be used throughout for electronic components. The use of selenium or other similar ageing devices are permitted only for voltage surge protection to other solid state components. Line voltage compensation shall be provided to minimize variations in the primary power supply.

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3.5 Performance. The welding machine, when connected to the specified power supply, shall produce a unidirectional current alternating between two levels. The pulsating arc current shall produce a spray transfer mode of welding in all welding positions and control ranges. The machines shall furnish an arc characteristic suitable for welding ferrous and nonferrous metals in the spray transfer mode.

3.6 Diagram of connections. A diagram of electrical connections shall be attached to the inside of the welding machines, and shall be coated with transparent varnish or a plastic covering.

3.7 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.8 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each welding machine. The nameplate shall contain the information listed below. If the welding machine is a special model, the model designation shall include the model of the basic standard welding machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer' s name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase frequency)
Contract Number or Order Number
National Stock Number (NSN) or Plant Equipment Code (NSN preferred)
Gross Weight
Date of Manufacture

3.9 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract.

3.10 Workmanship. Workmanship of the welding machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the

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specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1) quality conformance inspection shall consist of at least the examination in 4.5 and the tests in 4.6. Failure of the item to pass the examination or tests shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined to determine compliance with the requirements of 3.2 throug"h 3.9.

4.6 Tests. Except as otherwise specified (see 6.2.1), all tests shall follow the procedures of NEMA EW-1, when applicable.

4.6.1 High potential test. The welding machine shall be subjected to a high potential test as described in Testing Standards of NEMA EW-1. When the first article inspection is required, this test shall be performed after the rough handling test in 4.6.2.

4.6.2 Rough handling test. A chain or cable shall be attached to the lifting device of the welding machine. The welding machine shall be raised and then dropped to fall freely a distance between 15 cm (6 inches) minimum and 25 cm (10 inches) maximum upon a solid concrete slab. The drop shall be repeated so that the impact is made on three of the lower edges of the machine. There shall be no damage to the welding machine as a result of these tests other than minor denting of the cited lower edges.

4.6.3 Weld performance test. Two weld beads, at least six inches long, shall be deposited in the vertical position on a steel plate. The spray transfer mode of welding shall be used. The beads shall be welded showing no evidence of difficulty to start or maintain the weld. The weld beads shall be free from spatter normally observed in the globular transfer mode of welding. Welding beads shall be deposited in each pulse mode of the machine and for variable pulse machines , weld beads shall be deposited at various average amperages and corresponding variable frequency pulsation.

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Failure to perform any of the above tests satisfactorily shall be cause for rejection of the welding machines.

4.6.4 Temperature rise-test. With machine at room temperature, the temperatures of the primary and secondary windings of the transformer shall be recorded. The welding machine shall be operated at rated load at 100 percent duty cycle until stable temperatures are obtained. The load tests are to be run with resistance loading. Temperature rise shall be in accordance with the requirements of 3.4.1.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D-3951. When required packaging, packing and special marking shall be accomplished in accordance with MIL-W-45562 as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The welding machines specified herein are intended for use with the gas shielded welding processes wherein the spray transfer mode of welding is required. The welding machines are used to produce a pulsating welding power.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required, etc. (see 1.2).

c. When first article is required for inspection and approval (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or S1), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or S1), state required system (see 3.2.1.1).

f. Dual calibrations (US and S1), if required (see 3.2.1.2).

g. Additional safety and health requirements, if required (see 3.2.4).

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h. Portability, if required (see 3.2.6).

i. Painting, if different (see 3.3.5).

j. Power supply, if different specify input voltage, frequency and phase (see 3.4.7).

k. Specify and fully describe optional equipment, if required (see 3.7).

l. Nameplate, if different (see 3.8).

m. First article inspection, if different (see 4.3).

n. Quality conformance inspection, if different (see 4.4).

o. Tests, if different (see 4.6).

p* Level of packaging, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. All data requirements such as, parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results shall be requested on a DD Form 1423, Contract Data Requirements List, and shall be furnished in accordance with each specified DD Form 1664, Data Item Description as shown on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment'* such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1.

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6.5 Changes from previous issue. Asterisk are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activity:
Army -AL
Navy -SH
Air Force - 84
DLA- GS

User Activities:
Navy - MC, CG
DLA - GS

Preparing Activity:
DLA-IP

Project Number:
3431-0157

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See instructions - Reverse Side)

1. OOCUMENT NUMBER: MIL-w-80214R

]2. OOCUMENT TITLE: Welding Machines, Arc, DC, Transformer-Rectifier, PulsatinR Current

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b. AOORESS (Street, Clfy, St.tr. ZIP COA)

1.TYPEOF ORGANIZATION (Morkone)

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b. WORK TELEPHONE NUM8ER (hckdcAma Code) - Optlotml

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18. oATEoFsuffMIsSION (VVMMDD)

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